A4 ER Models and Normal Forms

Relational Databases
RDB (MTD.ba VZ)
SS26 2026

Submission via Moodle **16 Points**

In this assignment, you will create an ER model and then derive tables from them with a focus on normal forms. Please be advised that the exercises build on each other.

Exercise 1 (8 points) *ER Diagram*

Draw an **ER diagram** for the following system.

- Identify all entities and relationships.
- Clearly indicate **primary keys**, **foreign keys**, and **attributes**.
- Add **cardinalities** (1:N, M:N, etc.) for each relationship.
- Use appropriate **notations**. You can use any tool or draw by hand, but be consistent with the notation!
- Please note that information in a (bracket) is only to give you examples and not something to be modeled.

You are building a database for an MMO game's **guild and territory conquest** system.

Each **Player** has a username and a current level. Players can join a **Guild**. Each guild has a name, a founding year, and a motto.

The game world is divided into **Territories**. Each territory is located in a specific **Biome** (e.g., "Frozen Tundra", "Scorched Wastes"). Each biome has a climate description and a primary resource that can be harvested there. Note that many territories can share the same biome, and the climate and primary resource belong to the biome, not to individual territories.

Guilds **control** territories by conquering them. Each territory can be controlled by at most one guild at a time, but a guild may control many territories.

Territories may **border** each other (for example, "Frostpeak Ridge" borders "Glacier Pass"). This relationship is symmetric and should be recorded in the database.

Players are **assigned** to manage territories on behalf of their guild. Each assignment has a specific **Rank** (e.g., Scout, Sergeant, Captain, Commander). A player can be assigned to multiple territories with different ranks, and a territory can have multiple players assigned to it with different ranks. For each assignment, the system tracks the player's **contribution points** and the **date** they were assigned. Note that the rank itself has a permissions level that applies globally to that rank — it does not vary per assignment.

Exercise 2 (8 points) *Normal Forms*

1. Based on your ER model, write the SQL `CREATE TABLE` statements for your schema.
    - Define **primary keys** and **foreign keys**.
    - Use appropriate data types (`INT`, `VARCHAR`, `ENUM`, etc.).
    - Ensure that your relational schema is in **Third Normal Form (3NF)**. This means:
        - All attributes must depend only on the primary key.
        - No transitive dependencies are allowed.
        - No partial dependencies on part of a composite key.
    - **Briefly explain the primary and foreign keys (if you had to split a table for 3NF)** you have chosen for each table.